

Evan Nelles Henderson | Curriculum Vitae

MSc Student in Physics and Astronomy · McMaster University

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Research Interests

Origins of Life – Planetary Science – Prebiotic Chemistry – Magma Ocean Processes
Coupled Interior-Atmosphere Evolution – Exoplanets – Early Earth History

Education

M.Sc. in Physics (Astrobiology Collaborative Program) **2025 – Present**
McMaster University *Hamilton, Canada*
Thesis: *Geophysical Pathways to Prebiotic Atmospheres (In Progress)*
Supervisor: Dr. Ralph Pudritz

B.Sc. in Physics (Minor Natural History) **2020 – 2024**
McGill University *Montreal, Canada*

Research Experience & Fieldwork

Graduate Researcher — Planetary Interiors and Atmospheres **2025 – Present**
McMaster University *Hamilton, Canada*
With: Dr. Ralph Pudritz & Dr. Tim Lichtenberg

- Modelling the coupled evolution of magma oceans and protoatmospheres on rocky planets using the [PROTEUS](#) framework, with the goal of identifying geophysical pathways to prebiotic atmospheres.
- Conducting parameter sweeps of volatile outgassing scenarios to characterise atmospheric composition during and after magma ocean solidification.
- Researching surface-atmosphere interactions, including volcanism, serpentinization, and carbonate formation.

A Bayesian Investigation of the Transit Method **2023**
McGill University *Montreal, Canada*
With: Dr. Adrian Liu

- Modelled exoplanet transit light curves using MCMC methods as part of a supervised undergraduate project.

Palaeontology Field Research **2023**
McGill University *Dinosaur Provincial Park, Canada*
With: Dr. Hans Larsson

- Three weeks of Cretaceous fossil excavation and field data collection in the badlands as part of a supervised field research course.

Science Communication Experience

Author and Editor **2026 – Present**
Astrobites

Science Communicator **2025 – Present**
W. J. McCallion Planetarium *Hamilton, Canada*

Science Communication Publications

- [A Meditation on the Chemistry of WASP-39b](#), *Astrobites*, Jul 20th, 2026.
- [These “Ice Giants” Have a Hot Secret](#), *Astrobites*, Jun 25th, 2026.
- [Admiring the Atmospheric Egg Shell of Sub-Neptune TOI-270d](#), *Astrobites*, Apr 6th, 2026.
- [Hang On To Your Water, Super-Earths!](#), *Astrobites*, Feb 17th, 2026.

Presentations

Talks

Canadian Astronomical Society’s Annual General Meeting (CASCA) **2026**
Université de Montréal - Campus MIL *Montreal, Canada*

Waterloo-McMaster-York Exoplanet Symposium **2026**
University of Waterloo *Waterloo, Canada*

Departmental Astronomy Journal Club **2026**
McMaster University *Hamilton, Canada*

McMaster-Waterloo Exoplanet Symposium **2026**
McMaster University *Hamilton, Canada*

Posters

Emerging Researchers in Exoplanet Science (ERES XI) **2026**
University of Michigan *Ann Arbor, USA*

Teaching Experience

Teaching Assistant **2025 – Present**
McMaster University *Hamilton, Canada*

- The Big Questions ([ASTRON 2B03](#)) Fall 2026
- Integrated Science ([ISCI 1A24 A](#)) Fall 2025 & Fall 2026
- Introductory Mechanics ([PHYS 1D03](#)) Fall 2025
- Explorations in Medical and Biological Physics ([BIOPHYS 2S03](#)) Winter 2026
- Integrated Science ([ISCI 1A24 B](#)) Winter 2026

Technical Skills

Languages: Python, MATLAB, R, Bash, HTML, L^AT_EX, NetLogo

Scientific: NumPy, Matplotlib, Jupyter, emcee, batman, [PROTEUS](#) (SPIDER, AGNI, FastChem)

Simulation: Climate modelling, prebiotic chemistry, transit light curves, protocell replication

Operational: Git, Linux, SSH, HPC, Anaconda

Awards & Scholarships

- **McMaster Graduate Entrance Scholarship** **2025**
- **Martlet Student Athlete Honour Roll (McGill Varsity Rowing Team)** **2023**
- **Scotiabank Scholarship (Scholarships Partners Canada)** **2020 – 2024**